Networking

Fundamentals

Complete CCNA 200-301 v2.0 Study Guide

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CHAPTER 1

Introduction to Networking

Networking is the practice of connecting computers and other devices so they can communicate, share resources, and exchange data. A network can be as small as two computers linked by a cable or as large as the internet — a global system connecting billions of devices.

Networks underpin every modern IT system: email, file sharing, cloud services, video conferencing, VoIP, and business applications all depend on a functioning network. As a System Administrator, networking knowledge is essential because almost every IT problem eventually touches the network layer.

Types of Networks

Type	Full Name	Scope	Example
LAN	Local Area Network	Single building / campus	Office floor, home
WAN	Wide Area Network	Cities, countries, globe	MPLS between offices, Internet
WLAN	Wireless LAN	Local, over radio waves	Wi-Fi office network
MAN	Metropolitan Area Network	A city or metro area	City-wide ISP network
PAN	Personal Area Network	Around a person	Bluetooth headset
SAN	Storage Area Network	Data centre	Fibre Channel storage

Network Topologies

Topology	Description	Pros	Cons
Star	All devices connect to a central switch	Easy to manage, fault isolation	Single point of failure at switch
Bus	All devices share one cable	Simple, low cost	Collision domain, one failure affects all
Ring	Devices connected in a loop	Predictable performance	One break disrupts entire ring
Mesh	Every device connects to every other	High redundancy	Expensive, complex cabling
Hybrid	Combination of topologies	Flexible design	Can be complex to troubleshoot

Network Architecture Tiers

Enterprise networks follow structured designs to ensure scalability and reliability:

- **2-Tier (Collapsed Core):** Access layer + Distribution/Core layer combined. Used in small-to-medium networks. Simpler and cheaper but less scalable.
- **3-Tier:** Access layer (connects end devices) + Distribution layer (routing, policies, aggregation) + Core layer (high-speed backbone). Used in large enterprise networks.
- **Spine-Leaf:** Modern data centre design. Every leaf switch connects to every spine switch. Predictable latency, easy to scale. Common in cloud and SDN environments.

- **SOHO (Small Office / Home Office):** Simple flat network — one router/switch, all devices on the same subnet. No VLANs, no redundancy.

Network Devices and Their Roles

Device	OSI Layer	Function
Hub	Layer 1	Repeats signal to all ports — creates one collision domain. Legacy only.
Switch	Layer 2	Forwards frames based on MAC address. Each port = separate collision domain.
Layer 3 Switch	Layer 2/3	Switches AND routes. Uses SVIs for inter-VLAN routing.
Router	Layer 3	Forwards packets based on IP address. Separates broadcast domains.
Access Point	Layer 1/2	Provides wireless connectivity. Connects to wired switch.
WLC	Layer 2/3	Wireless LAN Controller. Centrally manages lightweight APs.
NGFW / IPS	Layer 3–7	Next-gen firewall — deep packet inspection, IPS, URL filtering.
Server	Layer 7	Provides services: DNS, DHCP, file, email, web, authentication.

CHAPTER 2

Cables, Interfaces & Physical Layer

The Physical Layer (OSI Layer 1) defines the electrical, optical, and mechanical specifications for transmitting raw bits over a medium. Understanding cabling and interface errors is essential for troubleshooting connectivity problems — many network outages start here.

Copper Cable Types

Cable	Standard	Max Speed	Max Distance	Use
Cat 5e	UTP 100 MHz	1 Gbps	100 m	Basic office LAN
Cat 6	UTP 250 MHz	1 Gbps / 10 Gbps	100 m / 55 m	Standard enterprise LAN
Cat 6A	UTP 500 MHz	10 Gbps	100 m	High-speed LAN, PoE+
Cat 7	STP 600 MHz	10 Gbps	100 m	Shielded, data centres
Coaxial	—	Variable	Variable	Legacy cable broadband

Copper Wiring Pinouts

Straight-through cable: Both ends use the same pinout (T568A–T568A or T568B–T568B). Used to connect different device types: PC to switch, router to switch.

Crossover cable: One end T568A, other end T568B. Used to connect like devices: switch to switch, PC to PC, router to router. Modern switches use Auto-MDIX to detect and correct automatically, so crossover cables are rarely needed today.

Rollover (Console) cable: All pins reversed (pin 1 to pin 8, etc.). Used to connect a PC serial port (or USB-to-serial adapter) to the console port of a Cisco router or switch for out-of-band management.

Fibre Optic Cable

Type	Core Size	Light Source	Max Distance	Use
Multimode (MMF)	50 / 62.5 µm	LED / VCSEL	Up to 550 m	Short distances: within a building / campus
Single-mode (SMF)	9 µm	Laser	Up to 80 km+	Long distances: between buildings, WAN links

Fibre connectors: LC (most common in enterprise), SC, ST, MPO (for high-density/trunk cables). SFP (Small Form-factor Pluggable) modules in switches accept either copper or fibre transceivers — the switch hardware is the same; the transceiver defines the medium.

Interface Errors — Reading show interfaces

The `show interfaces [type number]` command is your first tool for diagnosing physical and data-link problems. Key counters to check:

Counter	What It Means	Likely Cause
Input errors	Total of all input error types	Physical layer problem
CRC errors	Frame received with bad checksum	Bad cable, connector, NIC, or duplex mismatch
Collisions	Frames collided on the wire	Half-duplex or hub; should be zero on full-duplex
Late collisions	Collisions after first 64 bytes	Cable too long, duplex mismatch
Runts	Frames shorter than 64 bytes	Collision or bad NIC
Giants	Frames larger than 1518 bytes	MTU misconfiguration or jumbo frames issue
Output drops	Frames dropped in output queue	Interface congested, QoS needed
Interface resets	Link cycled up/down	Flapping cable, keepalive failure

Duplex mismatch is one of the most common causes of poor network performance. One side is set to full-duplex, the other to half-duplex. Symptoms: high CRC errors and late collisions, poor throughput but the link stays up. Fix: set both sides to the same speed and duplex explicitly, or both to auto.

Speed mismatch: If both sides are set to different speeds (e.g., one to 100 Mbps, one to 1 Gbps), the interface will show as down/down. Fix: match speeds or set both to auto.

PoE — Power over Ethernet

Standard	Name	Max Power (per port)	Use Case
802.3af	PoE	15.4 W	IP phones, basic APs, cameras
802.3at	PoE+	30 W	Dual-radio APs, PTZ cameras
802.3bt	PoE++	60 W / 90 W	High-performance APs, video conferencing endpoints

A PoE switch detects whether a device is PoE-capable before supplying power. Non-PoE devices are not harmed. The switch's total PoE budget must be considered when planning deployments — a 24-port switch might have a 370 W budget, enough for 24 PoE devices at 15 W each but not for 24 PoE+ devices at 30 W.

CHAPTER 3

The OSI Model

The Open Systems Interconnection (OSI) model is a seven-layer framework that standardises how networking systems communicate. Every network troubleshooting session uses this model — you work from the bottom up to isolate which layer the problem is at.

The Seven Layers

Layer	Name	Function	Key Protocols / Examples
7	Application	End-user services	HTTP, HTTPS, FTP, DNS, SMTP, SSH
6	Presentation	Data format, encryption, compression	TLS/SSL, JPEG, ASCII
5	Session	Session establishment, management, termination	NetBIOS, RPC
4	Transport	End-to-end delivery, segmentation, error recovery	TCP, UDP, port numbers
3	Network	Logical addressing and routing	IP, ICMP, OSPF, BGP
2	Data Link	Node-to-node delivery, MAC addressing, framing	Ethernet, 802.11, VLANs
1	Physical	Bits on the wire — cables, signals, connectors	RJ-45, fibre, hubs

Data Encapsulation

As data travels down the OSI stack, each layer adds its own header — a process called encapsulation. The data unit name changes at each layer:

- Layers 5–7: **Data**
- Layer 4 (Transport): **Segment** (TCP) or **Datagram** (UDP)
- Layer 3 (Network): **Packet**
- Layer 2 (Data Link): **Frame**

- Layer 1 (Physical): **Bits**

At the destination, de-encapsulation strips each header as data moves up the stack.

OSI vs TCP/IP Model

TCP/IP Layer	OSI Layers	Key Protocols
Application	5, 6, 7	HTTP, DNS, SMTP, FTP, SSH
Transport	4	TCP, UDP
Internet	3	IP, ICMP, ARP
Network Access	1, 2	Ethernet, Wi-Fi

Troubleshooting with OSI — Bottom-Up

Layer 1: Is the cable connected? Are link lights showing? Test with a different cable.

Layer 2: Is the MAC address in the switch's CAM table? Correct VLAN? Any port security violations?

Layer 3: Valid IP address? Can it ping the default gateway? Any routing issues?

Layer 4: Is the correct port open? Is a firewall blocking TCP/UDP?

Layer 7: Is the application configured correctly? Is the service running?

This bottom-up method stops you spending an hour debugging DNS when the problem is a faulty cable.

CHAPTER 4

IP Addressing & Subnetting

An IP address uniquely identifies a device on a network. IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses. IP addressing is the foundation of all routing decisions.

IPv4 Address Structure & Classes

Class	First Octet Range	Default Mask	Usable Hosts	Use
A	1–126	255.0.0.0	16,777,214	Large organisations, ISPs
B	128–191	255.255.0.0	65,534	Medium organisations
C	192–223	255.255.255.0	254	Small networks
D	224–239	N/A	N/A	Multicast
E	240–255	N/A	N/A	Experimental

Private IP Ranges (RFC 1918)

Range	CIDR	Class
10.0.0.0 – 10.255.255.255	10.0.0.0/8	A

Range	CIDR	Class
172.16.0.0 – 172.31.255.255	172.16.0.0/12	B
192.168.0.0 – 192.168.255.255	192.168.0.0/16	C

Special Addresses

Address	Purpose
127.0.0.1	Loopback — tests local IP stack
169.254.0.0/16	APIPA — assigned when DHCP fails (link-local)
0.0.0.0	Default route (any network)
255.255.255.255	Limited broadcast — all hosts on local segment
x.x.x.0	Network address — not assignable to a host
x.x.x.255	Directed broadcast for that subnet

Subnetting Reference Table

CIDR	Subnet Mask	Hosts/Subnet	Subnets from /24
/24	255.255.255.0	254	1
/25	255.255.255.128	126	2
/26	255.255.255.192	62	4
/27	255.255.255.224	30	8
/28	255.255.255.240	14	16
/29	255.255.255.248	6	32
/30	255.255.255.252	2	64
/31	255.255.255.254	2 (point-to-point)	—
/32	255.255.255.255	1 (host route)	—

Subnetting — How It Works

Subnetting divides a larger network into smaller sub-networks. Key formulas:

- **Subnets = 2^n** (n = bits borrowed from host portion)
- **Usable hosts = 2^h - 2** (h = remaining host bits; subtract 2 for network and broadcast)

Example: Subnet 192.168.1.0/24 into /26 subnets.

- Bits borrowed: 2 (26 - 24 = 2), so 4 subnets
- Host bits remaining: 6, so 62 usable hosts per subnet
- Subnets: 192.168.1.0/26, 192.168.1.64/26, 192.168.1.128/26, 192.168.1.192/26

IPv6 Addressing

IPv6 uses 128-bit addresses written as eight groups of four hex digits: 2001:0db8:85a3:0000:0000:8a2e:0370:7334. Rules: consecutive groups of zeros can be replaced with :: (once per address), and leading zeros in each group can be dropped.

IPv6 Address Type	Prefix	Description
Global Unicast	2000::/3	Routable on the internet — like public IPv4
Link-Local	FE80::/10	Auto-configured on every interface — not routed
Loopback	::1/128	Equivalent of 127.0.0.1
Multicast	FF00::/8	One-to-many — replaces IPv4 broadcast
Unique Local	FC00::/7	Private IPv6 — like RFC 1918, not internet routable

IPv6 EUI-64 Address Formation

EUI-64 is a method for automatically generating the 64-bit host portion of an IPv6 address from a device's MAC address. Steps:

1. Take the 48-bit MAC address: 00:1A:2B:3C:4D:5E

2. Split it in half and insert FFFE in the middle: 001A:2BFF:FE3C:4D5E

3. Flip the 7th bit (Universal/Local bit) of the first byte: 00 → 02 (in binary: 00000000 → 00000010)

4. Result host portion: 021A:2BFF:FE3C:4D5E

5. Combine with the /64 prefix: 2001:db8::/64 → 2001:db8::021A:2BFF:FE3C:4D5E

EUI-64 is used with Stateless Address Autoconfiguration (SLAAC) — devices configure their own IPv6 address without a DHCPv6 server.

IPv6 Address Configuration Methods

Method	How It Works	DHCPv6 Server Needed?
Manual (Static)	Administrator assigns address	No
SLAAC	Device uses EUI-64 + Router Advertisement prefix	No
Stateless DHCPv6	SLAAC for address + DHCPv6 for DNS/other options	Yes (stateless)
Stateful DHCPv6	DHCPv6 assigns full address + options	Yes (stateful)

NDP — Neighbour Discovery Protocol

NDP replaces ARP in IPv6. It uses ICMPv6 messages and multicast — no broadcasts. Key NDP messages:

- **RS (Router Solicitation):** Host asks for a router's prefix and default gateway info
- **RA (Router Advertisement):** Router sends prefix, default gateway, flags for SLAAC/DHCPv6
- **NS (Neighbour Solicitation):** Equivalent of ARP Request — "Who has this IPv6 address?"
- **NA (Neighbour Advertisement):** Equivalent of ARP Reply — "I have that address, here is my MAC"

Neighbour cache = IPv6's equivalent of the ARP cache.

CHAPTER 5

TCP, UDP & Transport Layer

The Transport Layer (Layer 4) is responsible for end-to-end communication between applications. The two main protocols are TCP (reliable) and UDP (fast).

TCP — Three-Way Handshake

TCP establishes a connection before any data is sent:

1. **SYN** — client sends synchronise packet to server

2. **SYN-ACK** — server acknowledges and sends its own synchronise

3. **ACK** — client acknowledges, connection is established

TCP features: sequence numbers, acknowledgements, retransmission of lost segments, flow control (sliding window), and a four-way FIN/ACK teardown.

TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Guaranteed delivery	Best-effort, no retransmission
Order	Segments reordered if needed	No ordering guarantee
Speed	Slower (overhead)	Faster (low overhead)
Header size	20–60 bytes	8 bytes
Use cases	HTTP, SSH, FTP, SMTP	DNS, DHCP, VoIP, video streaming

Well-Known Port Numbers

Port	Protocol	Service
20/21	TCP	FTP (data / control)
22	TCP	SSH
23	TCP	Telnet (unencrypted — avoid)
25	TCP	SMTP (email sending)
53	TCP/UDP	DNS
67/68	UDP	DHCP (server / client)
69	UDP	TFTP
80	TCP	HTTP
110	TCP	POP3
143	TCP	IMAP
161/162	UDP	SNMP agent / trap
443	TCP	HTTPS

Port	Protocol	Service
445	TCP	SMB (Windows file sharing)
514	UDP	Syslog
3389	TCP	RDP
22	TCP	SCP / SFTP (over SSH)

CHAPTER 6

Ethernet & Switching (Layer 2)

Ethernet (IEEE 802.3) is the dominant wired LAN technology. Modern networks use full-duplex switched Ethernet, which eliminates collisions entirely.

MAC Addresses & CAM Table

A MAC address is a 48-bit hardware address (e.g., 00:1A:2B:3C:4D:5E). The first 24 bits are the OUI (manufacturer); the last 24 bits identify the device. Switches build a MAC address table (CAM table) by learning source MACs. Unknown destinations are flooded; known destinations are forwarded only to the correct port.

VLANs — Virtual Local Area Networks

A VLAN logically segments a physical switch into multiple broadcast domains. Benefits: security, reduced broadcast traffic, flexibility. VLANs are defined by IEEE 802.1Q — a 4-byte tag is inserted into the Ethernet frame to identify the VLAN.

Access, Trunk & Voice Ports

Port Type	Description	Use
Access port	Carries one VLAN; tag stripped	End devices: PCs, printers, cameras
Trunk port	Carries multiple VLANs using 802.1Q tags	Switch-to-switch, switch-to-router links
Voice VLAN	Access port with a separate VLAN for IP phone traffic	IP phones — voice on one VLAN, PC data on another
Native VLAN	Untagged traffic on a trunk (default VLAN 1)	Change to unused VLAN for security

Voice VLAN: An IP phone connects to the switch, and a PC connects to the passthrough port on the phone. The switch port is an access port for data (untagged) and uses a voice VLAN for the phone's traffic (tagged). Config: `switchport voice vlan [ID]`

CDP and LLDP

CDP (Cisco Discovery Protocol) and LLDP (Link Layer Discovery Protocol) are Layer 2 neighbour discovery protocols. They advertise device identity, capabilities, and interface information to directly connected neighbours. They do NOT route — Layer 2 only.

CDP: Cisco proprietary. Enabled by default on Cisco devices. Shows: device ID, local/remote interface, platform, IOS version, IP address.

LLDP: IEEE 802.1AB — open standard, works between any vendor. Must be enabled manually on Cisco IOS (`lldp run`).

Use cases: verifying network documentation (confirm what is physically connected where), discovering neighbour IP addresses for troubleshooting, mapping the network topology.

Key commands:

- `show cdp neighbors detail` — see neighbour device details including IP
- `show lldp neighbors detail` — same for LLDP
- `no cdp run` — disable CDP globally (security best practice on edge ports)

Spanning Tree Protocol (STP / RSTP)

STP (IEEE 802.1D) and Rapid STP (802.1w) prevent Layer 2 loops on networks with redundant switch links.

STP Election Process:

1. Elect Root Bridge — lowest Bridge ID (priority + MAC). Default priority 32768. Lower = preferred. Set manually: `spanning-tree vlan [id] priority [value]`

2. Each non-root switch selects a Root Port — lowest-cost path to root

3. Each segment has one Designated Port (forwards). All others are Alternate (blocked)

Port States (RSTP): Discarding → Learning → Forwarding (Rapid STP converges in < 1 second vs 30–50 seconds for classic STP)

Key STP Features:

Feature	Purpose	Where to Apply
PortFast	Skip Listening/Learning — instant forwarding	Access ports connected to end devices ONLY
BPDU Guard	Err-disable port if BPDU received on PortFast port	All PortFast ports — prevents rogue switches
Root Guard	Prevent a port from becoming a root port	Ports where the root bridge should never be
Loop Guard	Prevent a blocked port from becoming designated if BPDUs stop	Redundant uplink ports
BPDU Filter	Stop sending/receiving BPDUs on a port	Edge ports — use carefully, can cause loops

EtherChannel (LACP)

EtherChannel bundles 2–8 physical links into one logical link — providing bandwidth aggregation and redundancy without STP blocking the extra links.

Protocols:

- **LACP (802.3ad):** IEEE open standard. Modes: Active (initiates) and Passive (responds). At least one side must be Active.
- **PAgP:** Cisco proprietary. Modes: Desirable (initiates) and Auto (responds).
- **Static (on):** No negotiation protocol. Both sides must be set to `on`.

All member ports must have identical speed, duplex, VLAN configuration, and trunk/access mode. Config: `interface port-channel [number]` → apply trunk/VLAN settings → then `channel-group [number] mode active` on each physical interface.

CHAPTER 7

Routing & Layer 3

Routing forwards packets from one network to another. A router examines the destination IP, performs a longest-prefix match in its routing table, and forwards the packet to the next hop.

Routing Table & Longest Prefix Match

The routing table contains entries from connected interfaces, static routes, and dynamic routing protocols. When a packet arrives, the router finds the most specific matching route (longest prefix). If no match exists and a default route is configured (0.0.0.0/0), the packet follows that. Otherwise the packet is dropped and an ICMP Unreachable message is sent back.

Administrative Distance

Route Source	AD	Preference
Connected interface	0	Most preferred
Static route	1	
eBGP	20	
EIGRP (internal)	90	
OSPF	110	
RIP	120	
EIGRP (external)	170	
iBGP	200	Least preferred (dynamic)

Static Routing

Static routes are manually configured. Types tested on the CCNA:

- **Network route:** `ip route 10.1.2.0 255.255.255.0 10.0.0.1` — matches a specific network
- **Default route:** `ip route 0.0.0.0 0.0.0.0 10.0.0.1` — matches anything when no specific route exists
- **Host route:** `ip route 10.1.2.5 255.255.255.255 10.0.0.1` — /32, matches exactly one IP
- **Floating static:** Same destination as a dynamic route but with a higher AD (e.g., AD 200). Acts as a backup — used only when the dynamic route disappears. Example: `ip route 10.1.2.0 255.255.255.0 10.0.0.2 200`

OSPFv2 — Configuration & Concepts

OSPF is a link-state routing protocol. Each router builds a complete topology map (LSDB) and runs Dijkstra's SPF algorithm. CCNA focuses on single-area OSPFv2 (for IPv4) and OSPFv3 (for IPv6).

Key OSPFv2 configuration:

`router ospf [process-id]` → `network [address] [wildcard] area [area-id]` → `router-id [A.B.C.D]`

Neighbour Requirements: Same area, same subnet, same hello/dead timers, same MTU, no authentication mismatch, same stub area flags.

Neighbour States: Down → Init → 2-Way → Exstart → Exchange → Loading → **Full**

DR/BDR Election (on multi-access networks like Ethernet): Highest OSPF priority wins (default 1; range 0–255; 0 = never DR). Tie-breaker: highest Router ID. DR reduces OSPF traffic — all routers form adjacency only with DR and BDR.

OSPF Cost: Reference bandwidth / Interface bandwidth. Default reference: 100 Mbps. A 100 Mbps link = cost 1; a 10 Mbps link = cost 10. Change with `auto-cost reference-bandwidth [Mbps]` to account for GigabitEthernet and above.

OSPFv3 for IPv6

OSPFv3 is OSPF for IPv6. The configuration is similar but uses interface-level commands instead of the `network` statement:

`ipv6 router ospf [process-id]` → on each interface: `ipv6 ospf [process-id] area [area-id]`

OSPFv3 uses IPv6 link-local addresses for neighbour relationships and runs over IPv6 multicast addresses FF02::5 (All OSPF Routers) and FF02::6 (All DR/BDR).

First Hop Redundancy — HSRP & VRRP

End devices are configured with a single default gateway IP. If that router fails, all traffic stops. First Hop Redundancy Protocols (FHRPs) solve this by creating a virtual IP that multiple routers share.

Protocol	Standard	Terms	Virtual MAC	Notes
HSRP	Cisco proprietary	Active / Standby	0000.0C07.ACxx	Two versions: HSRPv1 and HSRPv2
VRRP	IEEE 802.11r (open)	Master / Backup	0000.5E00.01xx	Open standard; actual router IP can be virtual IP
GLBP	Cisco proprietary	AVG / AVF	Multiple MACs	Load balances across multiple routers

How HSRP works:

- Both routers are configured with the same virtual IP (the default gateway for end devices)
- The Active router forwards traffic; the Standby monitors the Active
- If the Active fails, the Standby becomes Active within seconds (hello: 3 sec, hold: 10 sec)
- Priority determines which router is Active (default 100; higher wins). Configure preemption so the higher-priority router reclaims Active role when it recovers: `standby [group] preempt`

VRRP: Works the same way. The Master forwards; Backup routers monitor. The router with the highest priority (default 100) becomes Master. VRRP allows the virtual IP to be the same as the real interface IP of the Master router.

Inter-VLAN Routing

Method	How It Works	Pros	Cons
Router-on-a-Stick	One physical link, sub-interfaces per VLAN (802.1Q trunk)	Simple, uses existing router	Single point of congestion
Layer 3 Switch (SVI)	Switch Virtual Interfaces — IP on each VLAN interface	Hardware routing, fast, scalable	More expensive than L2 switch
Separate physical links	One router interface per VLAN	Simple, no tagging	Wastes interfaces, doesn't scale

NAT — Network Address Translation

Type	Description	Use Case
Static NAT	One private IP ↔ one fixed public IP	Servers that must be reachable from internet
Dynamic NAT	Pool of public IPs shared among hosts	Multiple hosts, unique public IPs
PAT / NAT Overload	Many private IPs share one public IP via port numbers	Home/office internet (most common)

CHAPTER 8

DHCP & DNS

DHCP — Dynamic Host Configuration Protocol

DHCP automatically assigns IP configuration to hosts. The four-step process is called **DORA:**

1. **Discover** — client broadcasts (UDP, 0.0.0.0 → 255.255.255.255, port 67/68)

2. **Offer** — DHCP server offers an IP address

3. **Request** — client broadcasts acceptance of the offer

4. **Acknowledge** — server confirms the lease

DHCP provides: IP address, subnet mask, default gateway, DNS server(s), lease duration.

DHCP Relay Agent: When the client and server are on different subnets, a router configured with `ip helper-address [server-IP]` converts the broadcast into a unicast to the DHCP server.

Troubleshooting DHCP:

Symptom	Likely Cause	Fix
169.254.x.x address	DHCP failed — APIPA assigned	Check DHCP server reachability, scope, relay
IP conflict error	Duplicate IPs	Exclude static IPs from DHCP scope
Wrong default gateway	Bad option in DHCP scope	Correct option 3 in DHCP server config
No IPs assigned	Scope exhausted	Shorten lease, expand pool, or find stale leases
DHCP works on same subnet, not remote	No relay agent	Configure ip helper-address on SVI or interface

DNS — Domain Name System

DNS translates hostnames to IP addresses. It is a distributed, hierarchical database. Resolution order: local cache → hosts file → recursive resolver → root server → TLD server → authoritative server.

DNS Record Types

Record	Purpose	Example
A	Hostname → IPv4 address	server1.co.com → 10.1.1.10
AAAA	Hostname → IPv6 address	server1.co.com → 2001:db8::1
CNAME	Alias → another hostname	www → server1.co.com

Record	Purpose	Example
MX	Mail exchanger for a domain	co.com → mail.co.com (priority 10)
PTR	Reverse lookup: IP → hostname	10.1.1.10 → server1.co.com
NS	Authoritative name server for domain	co.com NS ns1.co.com
SOA	Zone authority: serial, refresh, TTL	Zone management info
TXT	Arbitrary text: SPF, DKIM, verification	v=spf1 include:...

Diagnosing DNS Issues

Common DNS troubleshooting commands:

- `nslookup [hostname]` — query DNS and see which server responded
- `ping [hostname]` — if ping by IP works but not by name, it's a DNS problem
- `ipconfig /displaydns` (Windows) — view local DNS cache
- `ipconfig /flushdns` (Windows) — clear local DNS cache

Common issues: wrong DNS server configured (check DHCP option 6), record missing or pointing to wrong IP (check authoritative server), TTL too long (old IP cached after a change).

CHAPTER 9

Network Security Fundamentals

Network security protects the availability, integrity, and confidentiality of data. CCNA 200-301 covers the foundational concepts and controls you configure every day.

Security Concepts

Concept	Definition
Vulnerability	A weakness that can be exploited
Threat	A potential danger that could exploit a vulnerability
Risk	Likelihood × Impact of a threat exploiting a vulnerability
Exploit	Code or method that takes advantage of a vulnerability
Mitigation	Control or action that reduces risk
CIA Triad	Confidentiality, Integrity, Availability — the three security goals

Common Attack Types

Attack	Description	Mitigation
DoS / DDoS	Flood traffic to exhaust resources	Rate limiting, ACLs, upstream filtering
Spoofing	Faking source IP or MAC address	DAI, IP Source Guard, uRPF
Man-in-the-Middle	Intercepting traffic between two parties	Encryption (TLS), DAI, 802.1X
VLAN Hopping	Attacker sends double-tagged frames to reach another VLAN	Change native VLAN, disable DTP

Attack	Description	Mitigation
Rogue DHCP	Attacker runs fake DHCP server to redirect traffic	DHCP Snooping
ARP Poisoning	Attacker sends fake ARP replies to redirect traffic	Dynamic ARP Inspection (DAI)
Brute Force	Repeated login attempts to guess passwords	Account lockout, MFA, SSH with keys

ACLs — Access Control Lists

ACLs are ordered permit/deny rules applied to a router interface. There is an implicit **deny all** at the end of every ACL. Rules are matched top-to-bottom; first match wins.

Standard ACLs (1–99, 1300–1999): Match source IP only. Place **close to the destination**.

Extended ACLs (100–199, 2000–2699): Match source IP, destination IP, protocol, port numbers. Place **close to the source** to stop unwanted traffic early.

Named ACLs: Same function, easier to manage and edit lines by name.

Example extended ACL denying Telnet from 10.1.1.0/24 to any:

`ip access-list extended BLOCK-TELNET`

`deny tcp 10.1.1.0 0.0.0.255 any eq 23`

`permit ip any any`

Apply: `interface [type/num]` → `ip access-group BLOCK-TELNET in`

AAA — Authentication, Authorisation & Accounting

Feature	RADIUS	TACACS+
Standard	Open (RFC 2865/2866)	Cisco proprietary
Transport	UDP 1812/1813	TCP port 49
Encryption	Password only	Entire packet body
Auth + Author	Combined in one exchange	Separated (more granular control)
Best for	Network access: 802.1X, VPN, Wi-Fi	Device administration: CLI, exec commands

Layer 2 Security Features

DHCP Snooping: Switch acts as a firewall — only trusted ports (toward the real DHCP server) can send DHCP Offers and ACKs. Builds a binding table (MAC, IP, port) used by DAI and IP Source Guard.

Config: `ip dhcp snooping` (global) + `ip dhcp snooping vlan [id]` + `ip dhcp snooping trust` on uplink port.

Dynamic ARP Inspection (DAI): Validates ARP packets against the DHCP snooping binding table. Drops ARP replies where the MAC-to-IP mapping doesn't match the binding table. Prevents ARP spoofing/poisoning.

Config: `ip arp inspection vlan [id]` + `ip arp inspection trust` on uplink.

Port Security: Limits which MAC addresses can use a switch port. Can restrict the number of MACs or specify exact MACs. Violation modes: Protect (drop silently), Restrict (drop + log), Shutdown (err-disable port).

Config: `switchport port-security` + `switchport port-security maximum [n]` + `switchport port-security violation shutdown`

Storm Control: Limits the rate of broadcast, multicast, or unknown unicast traffic on a port. Prevents broadcast storms from consuming all bandwidth.

Config: `storm-control broadcast level [rising%] [falling%]`

RA Guard: Blocks rogue IPv6 Router Advertisement messages on ports where routers should not exist (access ports). Prevents a host from acting as a rogue IPv6 default gateway.

Config: `ipv6 nd raguard policy [name]` → `device-role host` → apply to interface.

802.1X Port-Based Authentication: Devices must authenticate (via RADIUS) before the switch port is opened. Roles: Supplicant (client), Authenticator (switch), Authentication Server (RADIUS).

VPNs & IPsec

A VPN creates an encrypted tunnel over an untrusted network. CCNA covers the concepts:

Site-to-Site VPN: Connects two networks permanently over the internet using IPsec. Both ends have a VPN gateway (firewall or router). Transparent to end users.

Remote Access VPN (Client VPN): Individual users connect securely from anywhere. Common protocols: SSL/TLS (AnyConnect), IPsec IKEv2.

IPsec Protocols:

Protocol	Full Name	Provides	Notes
AH	Authentication Header	Integrity + Authentication	No encryption — rarely used alone
ESP	Encapsulating Security Payload	Encryption + Integrity + Auth	Most common — provides full protection
IKE	Internet Key Exchange	Key negotiation	Phase 1: establish secure channel; Phase 2: negotiate IPsec SA

Mode	What Is Encrypted	Use Case
Transport mode	Payload only — original IP header intact	Host-to-host VPN
Tunnel mode	Entire original packet + new IP header added	Site-to-site and remote access VPN

Device Hardening

Key IOS hardening steps (also tested in CCNA):

- Set strong enable secret: `enable secret [password]`
- Encrypt all passwords: `service password-encryption`
- Login banners: `banner motd # Authorised users only #`
- Console security: `line console 0` → `login local` → `exec-timeout 5 0`
- VTY security (SSH only): `line vty 0 15` → `transport input ssh` → `login local`
- Generate SSH keys: `ip domain-name co.local` → `crypto key generate rsa modulus 2048` → `ip ssh version 2`
- Disable unused services: `no cdp run` on edge ports, `no ip http server`
- Use ACLs to restrict management access to the device

SFTP & SCP — Secure File Transfer

The CCNA exam requires you to know how to securely transfer IOS images and configuration files. TFTP (UDP 69) is unencrypted — do not use over untrusted networks.

SCP (Secure Copy Protocol): Uses SSH for encrypted file transfer.

- Copy to router: `copy scp://user@server/path/file.bin flash:`
- Copy from router: `copy flash:file.bin scp://user@server/path/`

SFTP (SSH File Transfer Protocol): Also SSH-based. Used with `ip sftp` commands or management platforms.

Use `ip scp server enable` on the IOS device to allow SCP transfers to it.

CHAPTER 10

IP Services — QoS, NTP, SNMP & Syslog

QoS — Quality of Service

QoS mechanisms ensure that delay-sensitive traffic (voice, video) gets priority over bulk data (file downloads, backups) on congested links.

QoS Mechanism	Description
Classification	Identify traffic type (by ACL, DSCP, CoS, NBAR)
Marking	Tag traffic with DSCP (Layer 3) or CoS/802.1p (Layer 2) value
Queuing	Multiple queues — high-priority traffic serviced first (LLQ, CBWFQ)
Shaping	Delay excess traffic to conform to a rate (smooth, adds delay)
Policing	Drop or re-mark traffic that exceeds a rate (hard limit)
Congestion avoidance	WRED — drop lower-priority packets early to prevent queue overflow

DSCP Values (common):

- **EF (Expedited Forwarding, DSCP 46):** Voice — the gold standard. Lowest latency, lowest jitter.
- **AF (Assured Forwarding, DSCP 34/36/26/28/18/20/10/12):** Video and business apps. Relative priority.
- **CS (Class Selector):** Backward compatible with IP Precedence (CS0–CS7).
- **BE (Best Effort, DSCP 0):** Default — no priority guarantee.

Trust boundary: The point in the network where you trust QoS markings from devices. Trust the switch port of an IP phone; do not trust markings from a PC (it can mark itself as EF).

Voice QoS requirements: Delay < 150 ms end-to-end, jitter < 30 ms, packet loss < 1%. Always put voice on a separate VLAN and queue with LLQ (Low-Latency Queuing).

NTP — Network Time Protocol

NTP synchronises device clocks across the network. Accurate time is critical for log correlation, certificate validity, Kerberos authentication (requires clocks within 5 minutes), and scheduled tasks. Uses UDP port 123.

NTP hierarchy: Stratum 0 = atomic clock / GPS reference. Stratum 1 = server directly connected to Stratum 0. Stratum 2 = synced to Stratum 1. Lower stratum number = more accurate.

Config: `ntp server [IP-of-NTP-server]`

Verify: `show ntp status` → `show ntp associations`

SNMP — Simple Network Management Protocol

SNMP monitors and manages network devices. Components: NMS (manager software), Agent (runs on each device), MIB (database of manageable variables / OIDs).

SNMP versions:

Version	Authentication	Encryption	Recommendation
SNMPv1	Community string (clear text)	None	Legacy — avoid
SNMPv2c	Community string (clear text)	None	Still common but weak
SNMPv3	Username + MD5/SHA	DES / AES	Required in production

SNMP operations: **GET** (read a value), **SET** (change a value), **TRAP** (unsolicited alert from device to NMS — UDP 162), **INFORM** (acknowledged trap — NMS sends back acknowledgement).

Syslog

Severity	Level	Keyword	Example Event
Emergency	0	emerg	System unusable — kernel panic
Alert	1	alert	Action needed immediately — temperature critical
Critical	2	crit	Critical hardware failure
Error	3	err	Interface error, process crash
Warning	4	warning	Link down, config change warning
Notice	5	notice	Interface up, normal significant event
Informational	6	info	Neighbour adjacency formed, ACL match logged
Debug	7	debug	Detailed protocol messages — very verbose

Config: `logging host [syslog-server-IP]` + `logging trap [severity-level]` + `service timestamps log datetime msec`

Verify: `show logging` (view buffer on the device itself)

CHAPTER 11

Wireless Networking (WLAN)

Wireless networking connects devices to a network without physical cables. Governed primarily by the IEEE 802.11 family of standards.

802.11 Standards

Standard	Frequency	Max Speed	Wi-Fi Generation	Key Feature
802.11b	2.4 GHz	11 Mbps	Wi-Fi 1	Legacy
802.11a	5 GHz	54 Mbps	Wi-Fi 2	Less interference, less range
802.11g	2.4 GHz	54 Mbps	Wi-Fi 3	Backward compatible with 802.11b
802.11n	2.4 / 5 GHz	600 Mbps	Wi-Fi 4	MIMO, dual-band
802.11ac	5 GHz	6.9 Gbps	Wi-Fi 5	MU-MIMO, wide channels
802.11ax	2.4 / 5 / 6 GHz	9.6 Gbps	Wi-Fi 6/6E	OFDMA, high density, low latency

RF Characteristics

Understanding RF (Radio Frequency) is essential for wireless design and troubleshooting:

RSSI (Received Signal Strength Indicator): Measures signal strength at the receiver. Expressed in dBm (negative values — closer to 0 = stronger). Good signal: > -70 dBm. Unusable: < -85 dBm.

SNR (Signal-to-Noise Ratio): Difference between signal strength and background noise. Higher SNR = better quality. SNR > 25 dB is generally good for voice/video.

Channel Planning (2.4 GHz): Only channels 1, 6, and 11 are non-overlapping. Adjacent APs must use different channels to avoid co-channel and adjacent-channel interference.

Channel Planning (5 GHz): Many non-overlapping channels available (24+) — much less congestion than 2.4 GHz.

Causes of interference:

- Co-channel interference: two APs on the same channel within range of each other
- Adjacent-channel interference: APs on overlapping channels (e.g., 1 and 3)
- Non-802.11 interference: microwave ovens, Bluetooth, cordless phones (all 2.4 GHz)
- Physical obstacles: concrete walls, metal racks absorb and reflect RF signals (5 GHz attenuates more than 2.4 GHz through walls)

Wireless Architecture

Architecture	Description	Use Case
Autonomous AP	Each AP independently configured	Small networks / temporary deployments
Lightweight AP + WLC	APs managed centrally via WLC (CAPWAP tunnel)	Enterprise — consistent config, roaming, RF management

Architecture	Description	Use Case
Cloud-based (Meraki)	APs managed via cloud dashboard	Distributed sites, easy management

CAPWAP (Control and Provisioning of Wireless Access Points): The protocol used between lightweight APs and a WLC. Uses UDP ports 5246 (control) and 5247 (data). All config is pushed from the WLC — the AP is 'dumb'. Roaming is seamless because the WLC maintains the session.

Wireless Security

Protocol	Authentication	Encryption	Status
WEP	Shared key (RC4)	RC4 (weak)	Broken — never use
WPA	PSK or 802.1X	TKIP	Deprecated
WPA2-Personal	PSK (passphrase)	AES-CCMP	Good — standard for home/SMB
WPA2-Enterprise	802.1X + RADIUS	AES-CCMP	Strong — individual user credentials
WPA3-Personal	SAE (replaces PSK)	AES-GCMP	Best current — forward secrecy
WPA3-Enterprise	802.1X + RADIUS	AES-256-GCMP	Strongest — required for Wi-Fi 6

Wireless Client Troubleshooting

When a wireless client can't connect or has poor performance, work through these steps:

Can't associate to SSID:

- Check SSID is correct and broadcasting (or hidden SSID — enter manually)
- Verify security protocol match (WPA2 vs WPA3)
- Check passphrase / credentials

Associated but no IP address:

- DHCP issue — check the DHCP server scope for the wireless VLAN
- Correct VLAN assigned to the SSID on the WLC?

IP address but can't reach resources:

- Check default gateway matches the wireless VLAN's SVI
- Check firewall/ACL between wireless VLAN and resources

Poor performance:

- Check RSSI (signal strength) — move closer or add AP
- Check SNR — look for interference sources
- Check channel utilisation — change channel or bandwidth
- Check if 2.4 GHz congested — move to 5 GHz

CHAPTER 12

Virtualisation & Cloud

The CCNA 200-301 exam tests fundamental virtualisation and cloud concepts as part of Domain 1. Understanding these is also essential for modern IT infrastructure.

Hypervisors

A hypervisor (Virtual Machine Monitor) allows multiple virtual machines (VMs) to run on a single physical host, each with its own OS and resources.

Type	Description	Examples	Use Case
Type 1 (Bare-metal)	Runs directly on hardware — no host OS	VMware ESXi, Microsoft Hyper-V, KVM	Data centres, production servers
Type 2 (Hosted)	Runs as an application on a host OS	VMware Workstation, VirtualBox, Parallels	Developer workstations, labs, testing

Containers

Containers package an application and its dependencies into a lightweight, portable unit that shares the host OS kernel — unlike VMs which each run a full OS.

- **Docker** is the most common container platform
- Containers start in seconds vs minutes for VMs
- Use less memory and storage than VMs
- **Kubernetes (K8s)** orchestrates containers at scale — manages deployment, scaling, and networking of containerised applications
- From a networking perspective, containers use virtual networks; each container gets an IP address within the container runtime's network bridge

Cloud Services & Deployment Models

Model	Description	Example
IaaS (Infrastructure as a Service)	Virtual machines, storage, networking in the cloud	Azure VMs, AWS EC2
PaaS (Platform as a Service)	Development platform — no OS management	Azure App Service, AWS Lambda
SaaS (Software as a Service)	Ready-to-use application over the internet	Microsoft 365, Salesforce
On-premises	All infrastructure owned and managed locally	Traditional data centre
Hybrid cloud	Mix of on-premises and cloud	Azure Arc, Entra ID + AD DS

CHAPTER 13

Network Automation & AI (Domain 5)

The CCNA v2.0 adds a new Domain 5 covering AI in networking and network automation. These topics carry 10% of the exam weight.

Network Management Approaches

Approach	Description	Examples
Device-based (Traditional)	Configure each device individually via CLI	Cisco IOS CLI, SSH sessions
Controller-based (SDN)	Centralised controller programs all devices via APIs	Cisco DNA Center, SD-WAN vManage
Cloud-based	Management plane in the cloud; devices phone home	Cisco Meraki, Cisco Umbrella
Automation-based	Scripts or tools push config changes to multiple devices	Ansible, Terraform, Python/Netmiko
Infrastructure as Code (IaC)	Network config stored in version-controlled code files	Terraform, Ansible playbooks in Git

SDN — Software-Defined Networking

Traditional networking: control plane (decides where traffic goes) and data plane (forwards traffic) run on the same device. SDN separates them:

- **Control plane** moves to a centralised **SDN Controller**
- **Data plane** (forwarding) remains on network devices
- Controller programs devices via southbound APIs (OpenFlow, NETCONF, RESTCONF)
- Applications talk to the controller via northbound APIs (REST API over HTTP/JSON)

Cisco DNA Center is Cisco's enterprise SDN controller. It provides intent-based networking — you define what you want (intent), and the controller translates that into device configs.

REST APIs

REST (Representational State Transfer) APIs allow software to interact with network devices and controllers programmatically. Uses HTTP methods:

HTTP Method	CRUD Action	Use in Networking
GET	Read	Retrieve interface status, routing table, device info
POST	Create	Create a new VLAN, add a user, deploy a new config
PUT	Update (replace)	Replace an entire config object
PATCH	Update (partial)	Change one field in a config object
DELETE	Delete	Remove a VLAN, revoke a policy

Data format: **JSON** (JavaScript Object Notation) is the most common. Human-readable, key-value pairs. **XML** is also used (older systems). API authentication: typically token-based (Bearer token) or Basic Auth (base64 username:password). API calls include headers: `Content-Type: application/json` and `Authorization: Bearer [token]`.

Ansible for Network Automation

Ansible is an agentless automation tool that uses SSH to connect to network devices and run commands or push configurations. No agent software needed on the device.

Key concepts:

- **Inventory:** List of devices Ansible manages (hosts file or YAML)
- **Playbook:** YAML file defining tasks to run on target devices
- **Module:** Reusable unit of work — e.g., `cisco.ios.ios_command` to run IOS commands, `cisco.ios.ios_config` to push config
- **Role:** Reusable collection of tasks, templates, and variables

Example playbook snippet:

`- name: Get IOS version`

` hosts: routers`

` tasks:`

` - name: Run show version`

` cisco.ios.ios_command:`

` commands: show version`

Ansible is idempotent — running the same playbook twice produces the same result without making unnecessary changes.

NETCONF & RESTCONF

NETCONF (RFC 6241): Uses XML over SSH (TCP 830) to configure network devices. Operates on YANG data models. More structured than CLI — designed for automation.

RESTCONF (RFC 8040): REST API version of NETCONF. Uses HTTP/HTTPS, JSON or XML, and YANG models. Easier to use with modern tools than NETCONF.

Both replace CLI for automated, programmatic network management at scale.

Agentic AI in Network Operations

CCNA v2.0 (2026) introduces AI concepts in networking — a new addition reflecting how AI tools are entering network operations:

Agentic AI: AI systems that can take autonomous actions to achieve a goal — not just answer questions but plan and execute multi-step tasks. In networking, this means AI that can: diagnose a network problem from logs, recommend or even apply a fix, interpret show command output, and generate configuration snippets.

Generative AI for network operations use cases:

- Analysing syslog output and identifying the root cause of an outage
- Generating Ansible playbooks or Python scripts from a plain-English description
- Summarising interface errors and recommending actions
- Explaining a routing table or ACL in plain language

Prompt engineering for network tasks: When using a generative AI system for network operations, a good prompt includes:

- **Persona:** "You are a Cisco network engineer..."
- **Context / Data:** Paste the relevant show command output or config
- **Task / Instructions:** "Analyse the following syslog messages and identify the issue"
- **Output format:** "Respond in bullet points" or "Provide the IOS commands to fix this"
- **Data classification note:** Never paste sensitive credentials or internal IP schemas into public AI tools

The exam may present AI-generated output or recommendations and ask you to evaluate whether they are correct.

CHAPTER 14

WAN Technologies

Technology	Type	Speed	Notes
Leased Line (T1/E1)	Dedicated point-to-point	1.5 / 2 Mbps	Expensive, guaranteed bandwidth — mostly legacy
MPLS	Provider-managed, packet-switched	Variable	Enterprise WAN standard — any-to-any, QoS classes
Metro Ethernet	Fibre Ethernet over MAN	10 Mbps – 10 Gbps	High-speed, low latency
Internet VPN (IPsec)	Public internet + encryption	Variable	Cost-effective; quality varies
SD-WAN	Software-defined, multi-link	Variable	Intelligent routing across MPLS + internet
4G / 5G Cellular	Mobile broadband	Variable	Backup or primary for remote sites
DSL	Over telephone lines	Up to 100 Mbps	Small offices, ADSL/VDSL
Cable	Over coaxial cable	Up to 1 Gbps	SMB; shared bandwidth with neighbourhood

MPLS — Multiprotocol Label Switching

MPLS assigns a label to each packet at the ingress PE router. Core LSR routers forward packets purely by label — faster than IP routing at each hop.

Key concepts: Label Edge Router (LER/PE) adds/removes labels. Label Switch Router (LSR/P) forwards by label only. MPLS VPN isolates customer traffic using VRFs. MPLS-TE provides traffic engineering. Most enterprise MPLS includes QoS classes (Gold/Silver/Bronze) guaranteeing latency/bandwidth for voice and critical apps.

PPP and HDLC

Legacy Layer 2 encapsulations for serial (point-to-point) WAN links — now largely replaced by Ethernet:

HDLC: Cisco default on serial interfaces. Cisco's version is proprietary — incompatible between vendors.

PPP: Open standard. Supports PAP authentication (clear-text — avoid) and CHAP (challenge-response, secure). Supports Multilink PPP (bonding multiple links) and compression.

CHAPTER 15

Cisco IOS — CLI Reference

IOS Exec Modes

Mode	Prompt	Access	Purpose
User EXEC	Router>	Default on login	Limited: ping, traceroute, basic show
Privileged EXEC	Router#	enable	Full show, debug, copy, reload
Global Config	Router(config)#	configure terminal	Change global settings
Interface Config	Router(config-if)#	interface [type num]	Configure interface
Line Config	Router(config-line)#	line console|vty [num]	Configure console / VTY
Router Config	Router(config-router)#	router ospf|eigrp|bgp	Configure routing protocol

Essential Show Commands

Command	What It Shows
show version	IOS version, uptime, hardware, license
show ip interface brief	All interfaces: IP, status (up/down), protocol
show interfaces [type num]	Detailed stats: errors, speed, duplex, counters
show ip route	Full routing table with source, AD, metric
show running-config	Active config in RAM
show startup-config	Config saved in NVRAM
show vlan brief	VLAN IDs, names, assigned ports
show mac address-table	Switch MAC-to-port mapping (CAM table)
show cdp neighbors detail	Directly connected Cisco device details
show lldp neighbors detail	Directly connected LLDP device details
show ip ospf neighbor	OSPF neighbour relationships and state
show ip ospf database	Link State Database — topology map
show standby	HSRP group status: Active/Standby, virtual IP
show vrrp	VRRP group status: Master/Backup
show access-lists	All ACL entries and hit counts
show crypto isakmp sa	IPsec Phase 1 SAs (VPN status)
show port-security	Port security status per interface
show ip dhcp binding	DHCP bindings: IP, MAC, lease time
show logging	Syslog buffer on the device

Key Troubleshooting Commands

Command	Purpose
ping [IP]	Test Layer 3 reachability (ICMP echo)
ping [IP] source [interface]	Ping from a specific source interface — useful for routing verification
traceroute [IP]	Hop-by-hop path to destination
show ip arp	ARP cache — MAC to IP mappings
debug ip ospf events	Real-time OSPF events (use carefully)
no debug all	Stop ALL debug output immediately
clear ip route *	Clear routing table (forces re-convergence)
clear arp-cache	Clear ARP table
terminal monitor	See log messages in a remote SSH/Telnet session
show etherchannel summary	EtherChannel status and member interfaces
show spanning-tree	STP topology: root bridge, port roles/states

CHAPTER 16

Quick Reference & Exam Tips

CCNA 200-301 v2.0 Exam Domain Weights

Domain	Topic	Weight
1.0	Network Infrastructure and Connectivity	25%
2.0	Switching and Network Access	25%
3.0	IP Routing	20%
4.0	Network Services and Security	20%
5.0	AI, and Network Operations and Management	10%

Key Numbers to Memorise

Value	What It Means
0	AD of a connected route
1	AD of a static route
20	AD of eBGP
90	AD of EIGRP internal
110	AD of OSPF
120	AD of RIP
200	AD of iBGP
15	Maximum RIP hop count

Value	What It Means
32768	Default STP bridge priority
100 Mbps	OSPF default reference bandwidth
224.0.0.5	OSPF All Routers multicast
224.0.0.6	OSPF DR / BDR multicast
FF02::5	OSPFv3 All Routers multicast (IPv6)
< 1 sec	RSTP convergence time
30–50 sec	Classic STP (802.1D) convergence
67/68	DHCP server/client (UDP)
123	NTP (UDP)
161/162	SNMP agent/trap (UDP)
5246/5247	CAPWAP control/data (UDP)
3 sec / 10 sec	HSRP hello / hold timers (default)

Common Exam Traps

- **ACL implicit deny:** Every ACL ends with an invisible `deny any` — if nothing matches, traffic is dropped. Always add a `permit ip any any` at the bottom if needed.
- **Standard ACL placement:** Close to the DESTINATION (standard ACLs match only source IP — placed close to source could block too much).
- **Extended ACL placement:** Close to the SOURCE (stop unwanted traffic early).
- **VLAN 1 is the default native VLAN** — change it on all trunk ports for security.
- **STP root bridge election:** Lowest Bridge ID wins. Lower priority number = more likely to become root. Set root bridge manually with lower priority (`spanning-tree vlan X priority 4096`).
- **OSPF Router ID:** Highest loopback IP → highest physical interface IP → manually set. Set it manually for stability.
- **OSPF neighbour won't form:** Check same area, same subnet, same hello/dead timers, same MTU, no authentication mismatch.
- **HSRP preemption:** Higher-priority router does NOT automatically take over when it returns — must configure `standby X preempt`.
- **EtherChannel:** All member ports must have identical speed, duplex, VLAN, and trunk mode. Mismatch causes the channel to fail.
- **PortFast + BPDU Guard:** PortFast on access ports ONLY. BPDU Guard shuts the port if a switch is plugged in.
- **NAT overload (PAT):** Thousands of private hosts share ONE public IP via unique source port numbers.
- **DHCP scope:** Always exclude the gateway IP (and any other static IPs) from the DHCP pool.
- **IPv6 EUI-64:** Remember to flip the 7th bit of the first byte of the MAC address.
- **CDP/LLDP:** Layer 2 only — do not cross routers unless specifically configured. Disable on edge (untrusted) ports.

Troubleshooting Methodology (CCNA Exam)

Follow this structured approach in exam simulation questions and real life:

1. **Define the problem** — what exactly is failing? Who is affected? When did it start?
2. **Gather information** — show commands, ping, traceroute, check logs
3. **Consider recent changes** — most outages follow a change

4. **Eliminate possibilities** — work bottom-up (OSI Layer 1 → 7)
5. **Identify root cause** — not just the symptom
6. **Implement one fix at a time** — one change, then test
7. **Verify** — test from the user's perspective, confirm fix
8. **Document** — what was wrong, what you changed, result